**Name:** Pirjade Sabahat Rehan
**ID:** COMETFWC056
**Date:** January 26, 2026

# RELATIONS AND FUNCTIONS

## EXERCISE 2.1

1. If $\left(\frac{x}{3}+1, y-\frac{2}{3}\right)=\left(\frac{5}{3}, \frac{1}{3}\right)$, find the values of $x$ and $y$.
2. If the set $A$ has 3 elements and the set $B=\{3,4,5\}$, then find the number of elements in $(A \times B)$. 3. If $G=\{7,8\}$ and $H=\{5,4,2\}$, find $G \times H$ and $H \times G$.
4. State whether each of the following statements are true or false. If the statement is false, rewrite the given statement correctly.

   (i) If $P=\{m,n\}$ and $Q=\{n,m\}$, then $P \times Q=\{(m,n),(n,m)\}$.

   (ii) If $A$ and $B$ are non-empty sets, then $A \times B$ is a non-empty set of ordered pairs $(x,y)$ such that $x \in A$ and $y \in B$.

   (iii) If $A=\{1,2\}$, $B=\{3,4\}$, then $A \times (B \cap \phi)=\phi$.

5. If $A=\{-1,1\}$, find $A \times A \times A$.
6. If $A \times B=\{(a,x),(a,y),(b,x),(b,y)\}$, find $A$ and $B$.
7. Let $A=\{1,2\}$, $B=\{1,2,3,4\}$, $C=\{5,6\}$ and $D=\{5,6,7,8\}$. Verify that:

   (i) $A \times (B \cap C)=(A \times B) \cap (A \times C)$

   (ii) $A \times C$ is a subset of $B \times D$.

8. Let $A=\{1,2\}$ and $B=\{3,4\}$. Write $A \times B$. How many subsets will $A \times B$ have? List them.
9. Let $A$ and $B$ be two sets such that $n(A)=3$ and $n(B)=2$. If $(x,1),(y,2),(z,1)$ are in $A \times B$, find $A$ and $B$.
10. The Cartesian product $A \times A$ has 9 elements among which are found $(-1,0)$ and $(0,1)$. Find the set $A$ and the remaining elements of $A \times A$.

## 2.3 Relations

Consider the two sets $P=\{a,b,c\}$ and $Q=\{\text{Ali, Bhanu, Binoy, Chandra, Divya}\}$. The Cartesian product of $P$ and $Q$ has 15 ordered pairs: $P \times Q=\{(a,\text{Ali}),(a,\text{Bhanu}),\ldots,(c,\text{Divya})\}$.

We can now obtain a subset of $P \times Q$ by introducing a relation $R$ as: $R=\{(x,y) : x \text{ is the first letter of the name } y, x \in P, y \in Q\}$.

Then $R=\{(a,\text{Ali}),(b,\text{Bhanu}),(b,\text{Binoy}),(c,\text{Chandra})\}$. An arrow diagram of this relation $R$ is shown in Fig. 2.4.

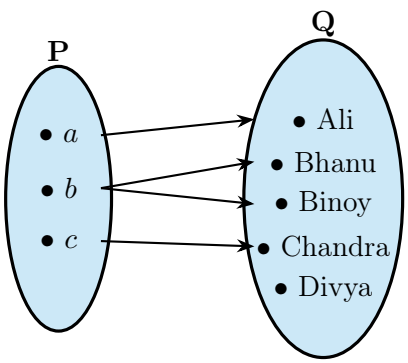



**Fig 2.4**

**Definition 2** A relation $R$ from a non-empty set $A$ to a non-empty set $B$ is a subset of the cartesian product $A \times B$. The subset is derived by describing a relationship between the first element and the second element of the ordered pairs in $A \times B$. The second element is called the *image* of the first element. **Definition 3** The set of all first elements of the or-

dered pairs in a relation $R$ from a set $A$ to a set $B$ is called the *domain* of the relation $R$.

**Definition 4** The set of all second elements in a relation $R$ from a set $A$ to a set $B$ is called the *range* of the relation $R$. The whole set $B$ is called the *codomain* of the relation $R$. Note that range $\subset$ codomain.

***Remarks***

1. A relation may be represented algebraically either by the *Roster method* or by the *Set-builder method.*

2. An arrow diagram is a visual representation of a relation.

**Example 7** Let $A = \{1, 2, 3, 4, 5, 6\}$. Define a relation $R$ from $A$ to $A$ by $R = \{(x, y) : y = x+1\}$

(i) Depict this relation using an arrow diagram.

(ii) Write down the domain, codomain and range of $R$.

**Solution** (i) By the definition of the relation,

$$R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6)\}.$$

The corresponding arrow diagram is shown in Fig. 2.5.

(ii) We can see that the domain $= \{1, 2, 3, 4, 5\}$. Similarly, the range $= \{2, 3, 4, 5, 6\}$ and the codomain $= \{1, 2, 3, 4, 5, 6\}$.

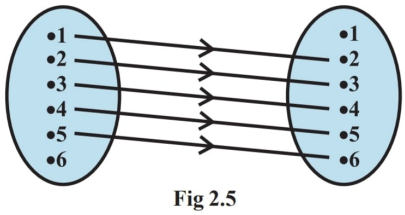


Fig 2.5

**Example 8** The Fig 2.6 shows a relation between the sets $P$ and $Q$. Write this relation

(i) in set-builder form,

(ii) in roster form.

What is its domain and range? **Solution** It is obvious that the relation $R$ is "$x$ is the square of $y$".

(i) **In set-builder form,**

$$R = \{(x, y) : x \text{ is the square of } y,\ x \in P,\ y \in Q\}$$

(ii) **In roster form,**

$$R = \{(9, 3), (9, -3), (4, 2), (4, -2), (25, 5), (25, -5)\}$$

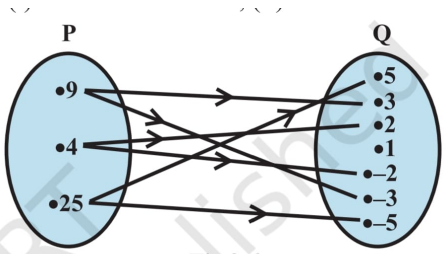


Fig 2.6

The domain of this relation is $\{4, 9, 25\}$.
The range of this relation is $\{-2, 2, -3, 3, -5, 5\}$.
Note that the element 1 is not related to any element in set $P$. The set $Q$ is the codomain of this relation.

**Note** The total number of relations that can be defined from a set $A$ to a set $B$ is the number of possible subsets of $A \times B$. If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$ and the total number of relations is $2^{pq}$.

**Example 9** Let $A = \{1, 2\}$ and $B = \{3, 4\}$. Find the number of relations from $A$ to $B$.

**Solution** We have,

$$A \times B = \{(1,3),(1,4),(2,3),(2,4)\}.$$

Since $n(A \times B) = 4$, the number of subsets of $A \times B$ is $2^4$. Therefore, the number of relations from $A$ into $B$ will be $2^4$. ***Remarks*** A relation $R$ from $A$ to $A$ is also stated as a relation on $A$.

## EXERCISE 2.2

1. Let $A = \{1, 2, 3, \ldots, 14\}$. Define a relation $R$ from $A$ to $A$ by $R = \{(x, y) : 3x - y = 0,$ where $x, y \in A\}$. Write down its domain, codomain and range.>

2. Define a relation $R$ on the set $\mathbb{N}$ of natural numbers by $R = \{(x, y) : y = x+5, x$ is a natural number less th $\mathbb{N}\}$. Depict this relationship using roster form. Write down the domain and the range.

3. $A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation $R$ from $A$ to $B$ by $R = \{(x, y) :$ the difference between $x$ and $y$ is odd; $x \in A, y \in B\}$. Write $R$ in roster form.

4. The Fig 2.7 shows a relationship between the sets $P$ and $Q$. Write this relation

   (i) in set-builder form

   (ii) in roster form.

   What is its domain and range?

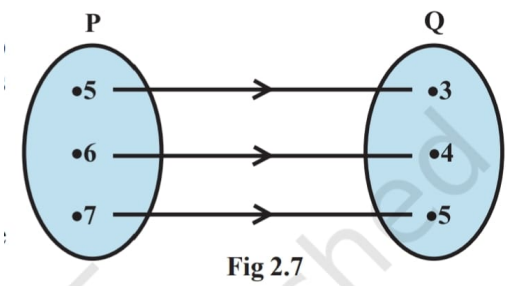


Fig 2.7

5. Let $A = \{1, 2, 3, 4, 6\}$. Let $R$ be the relation on $A$ defined by $\{(a, b) : a, b \in A, b$ is exactly divisible by $a\}$.

   (i) Write $R$ in roster form

   (ii) Find the domain of $R$

   (iii) Find the range of $R$.

6. Determine the domain and range of the relation $R$ defined by $R = \{(x, x + 5) : x \in \{0, 1, 2, 3, 4, 5\}\}$.

7. Write the relation $R = \{(x, x^3) : x$ is a prime number less than $10\}$ in roster form.

8. Let $A = \{x, y, z\}$ and $B = \{1, 2\}$. Find the number of relations from $A$ to $B$.

9. Let $R$ be the relation on $\mathbb{Z}$ defined by $R = \{(a, b) : a, b \in \mathbb{Z}, a - b$ is an integer$\}$. Find the domain and range of $R$.

**2.4 Functions** In this Section, we study a special type of relation called function. It is one of the most important concepts in mathematics. We can visualise a function as a rule, which produces new elements out of some given elements. There are many terms such as 'map' or 'mapping' used to denote a function.

**Definition 5** A relation $f$ from a set $A$ to a set $B$ is said to be a function if every element of set $A$ has one and only one image in set $B$. In other words, a function $f$ is a relation from a non-empty set $A$ to a non-empty set $B$ such that the domain of $f$ is $A$ and no two distinct ordered pairs in $f$ have the same first element.

If $f$ is a function from $A$ to $B$ and $(a, b) \in f$, then $f(a) = b$, where $b$ is called the **image** of $a$ under $f$ and $a$ is called the **preimage** of $b$ under $f$.

The function $f$ from $A$ to $B$ is denoted by $f : A \to B$.

Looking at the previous examples, we can easily see that the relation in Example 7 is not a function because the element 6 has no image.

Again, the relation in Example 8 is not a function because the elements in the domain are connected to more than one images. Similarly, the relation in Example 9 is also not a function. (Why?) In the examples given below, we will see many more relations some of which are functions and others are not. **Example 10** Let $\mathbb{N}$ be the set of natural numbers and the relation $R$ be defined on $\mathbb{N}$ such that $R = \{(x, y) : y = 2x, x, y \in \mathbb{N}\}$.
What is the domain, codomain and range of $R$? Is this relation a function? **Solution** The domain of $R$ is the set of natural numbers $\mathbb{N}$. The codomain is also $\mathbb{N}$. The range is the set of even natural numbers.
Since every natural number $n$ has one and only one image, this relation is a function. **Example 11** Consider the following relations. For each relation, determine whether it is a function or not, giving reasons.

$$\text{(i)} \quad R = \{(2, 1), (3, 1), (4, 2)\}, \ \text{(ii)} \quad R = \{(2, 2), (2, 4), (3, 3), (4, 4)\}$$

$$\text{(iii)} \quad R = \{(1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7)\}$$

**Solution** (i) Since 2, 3, 4 are the elements of domain of $R$ having their unique images, this relation $R$ is a function.
(ii) Since the same first element 2 corresponds to two different images 2 and 4, this relation is not a function.
(iii) Since every element has one and only one image, this relation is a function. **Definition 6**

A function which has either $\mathbb{R}$ or one of its subsets as its range is called a real valued function. Further, if its domain is also either $\mathbb{R}$ or a subset of $\mathbb{R}$, it is called a real function.
**Example 12** Let $\mathbb{N}$ be the set of natural numbers. Define a real valued function $f : \mathbb{N} \to \mathbb{N}$ by $f(x) = 2x + 1$. Using this definition, complete the table given below.

| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $y$ | $f(1) =$ | $f(2) =$ | $f(3) =$ | $f(4) =$ | $f(5) =$ | $f(6) =$ | $f(7) =$ |

**Solution** The completed table is given by:

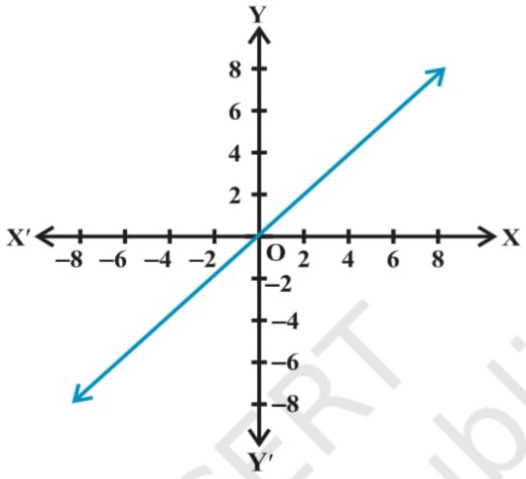
| $x$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| $y$ | $f(1) = 3$ | $f(2) = 5$ | $f(3) = 7$ | $f(4) = 9$ | $f(5) = 11$ | $f(6) = 13$ | $f(7) = 15$ |

### 2.4.1 Some functions and their graphs

**(i) Identity function** Let $\mathbb{R}$ be the set of real numbers. Define the real valued function $f : \mathbb{R} \to \mathbb{R}$ by

$$y = f(x) = x \quad \text{for each } x \in \mathbb{R}.$$

Such a function is called the *identity function*. Here, the domain and range of $f$ are $\mathbb{R}$. The graph is a straight line as shown in Fig 2.8. It passes through the origin.



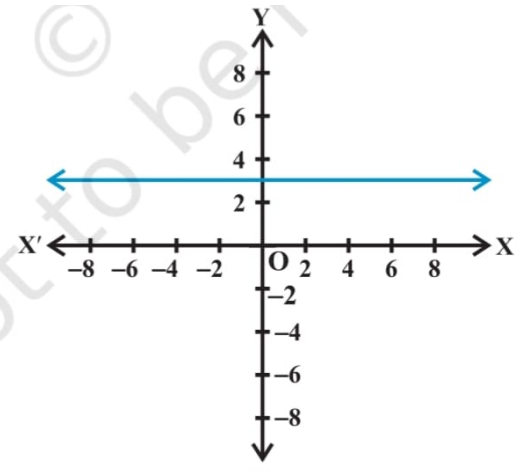

$f(x) = x$

**Fig 2.8**

**(ii) Constant function** Define the function $f : \mathbb{R} \to \mathbb{R}$ by

$$y = f(x) = c, \quad x \in \mathbb{R},$$

where $c$ is a constant. Here, the domain of $f$ is $\mathbb{R}$ and its range is $\{c\}$.




$f(x) = 3$

**Fig 2.9**

The graph is a line parallel to the $x$-axis. For example, if $f(x) = 3$ for each $x \in \mathbb{R}$, then its graph will be a line as shown in Fig. 2.9.

**(iii) Polynomial function** A function $f : \mathbb{R} \to \mathbb{R}$ is said to be a polynomial function if for each $x \in \mathbb{R}$,

$$y = f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n,$$

where $n$ is a non-negative integer and $a_0, a_1, a_2, \ldots, a_n \in \mathbb{R}$.

The functions defined by $f(x) = x^3 - x^2 + 2$ and $g(x) = x^4 + \dfrac{2}{x}$ are some examples of polynomial functions, whereas the function $h$ defined by

$$h(x) = \frac{x^{2/3}}{+} 2x$$

is not a polynomial function. (Why?)

**Example 13** Define the function $f : \mathbb{R} \to \mathbb{R}$ by $y = f(x) = x^2,\ x \in \mathbb{R}$. Complete the Table given below by using this definition. What is the domain and range of this function? Draw the graph of $f$.

| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|---|---|
| $y = f(x) = x^2$ | | | | | | | | | |

**Solution** The completed Table is given below:

| $x$ | $-4$ | $-3$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|---|---|---|
| $y = f(x) = x^2$ | 16 | 9 | 4 | 1 | 0 | 1 | 4 | 9 | 16 |

Domain of $f = \{x : x \in \mathbb{R}\}$. Range of $f = \{x^2 : x \in \mathbb{R}\}$. The graph of $f$ is given by Fig 2.10.

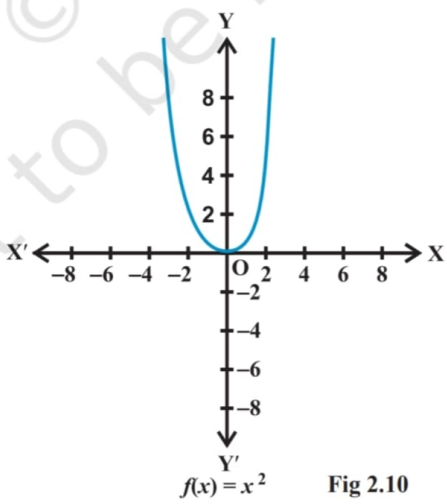


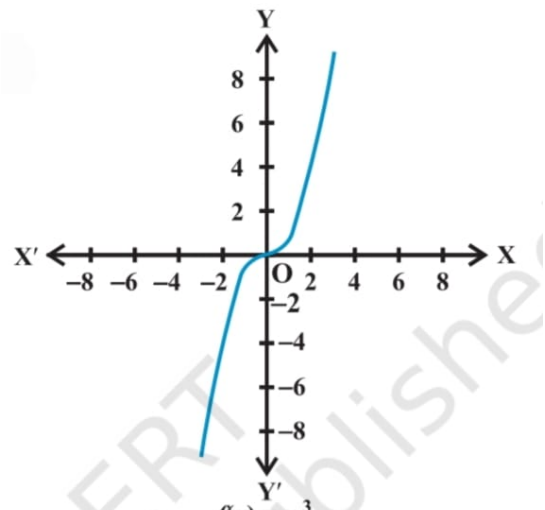

**Fig 2.10**

**Example 14** Draw the graph of the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^3,\ x \in \mathbb{R}$.

**Solution** We have $f(0) = 0,\ f(1) = 1,\ f(-1) = -1,\ f(2) = 8,\ f(-2) = -8,\ f(3) = 27,\ f(-3) = -27$, etc. Therefore,

$$f = \{(x, x^3) : x \in \mathbb{R}\}.$$

The graph of $f$ is given in Fig. 2.11.




$f(x) = x^3$

**Fig 2.11**

**Example 15** Define the real valued function

$$f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$$

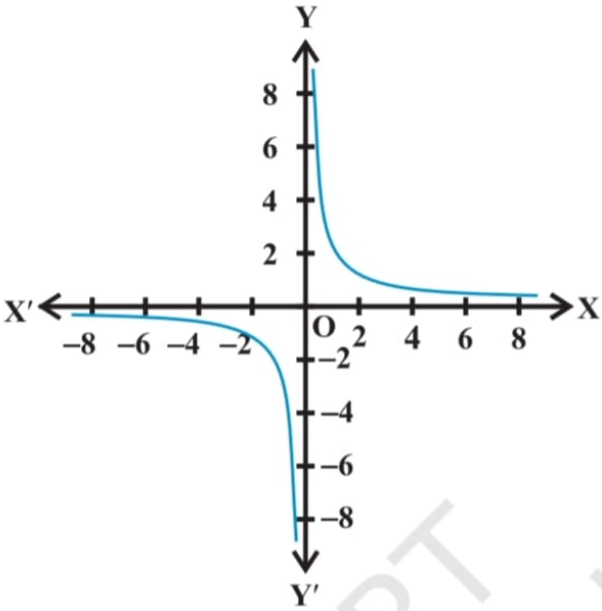
defined by $f(x) = \dfrac{1}{x}$, $x \in \mathbb{R} \setminus \{0\}$, Complete the Table given below using this definition. What is the domain and range of this function?

| $x$ | $-2$ | $-1.5$ | $-1$ | $-0.5$ | $0.25$ | $0.5$ | $1$ | $1.5$ | $2$ |
|---|---|---|---|---|---|---|---|---|---|
| $y = \dfrac{1}{x}$ | ... | ... | ... | ... | ... | ... | ... | ... | ... |

**Solution** The completed Table is given by

| $x$ | $-2$ | $-1.5$ | $-1$ | $-0.5$ | $0.25$ | $0.5$ | $1$ | $1.5$ | $2$ |
|---|---|---|---|---|---|---|---|---|---|
| $y = \dfrac{1}{x}$ | $-0.5$ | $-0.67$ | $-1$ | $-2$ | $4$ | $2$ | $1$ | $0.67$ | $0.5$ |

Domain of $f = \mathbb{R} \setminus \{0\}$. Range of $f = \mathbb{R} \setminus \{0\}$. The domain of $f$ is all real numbers except 0 and its range is also all real numbers except 0. The graph of $f$ is given in Fig 2.12.




$f(x) = \frac{1}{x}$

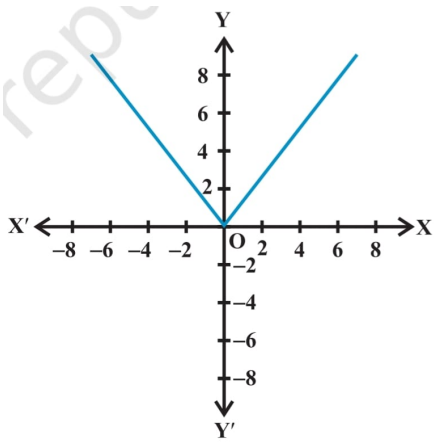
**Fig 2.12**

**(v) The Modulus function**

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = |x|$ for each $x \in \mathbb{R}$ is called the *modulus function*. For each non-negative value of $x$, $f(x)$ is equal to $x$. But for negative values of $x$, the value of $f(x)$ is the negative of the value of $x$, i.e.,

$$f(x) = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$

The graph of the modulus function is given in Fig. 2.13.



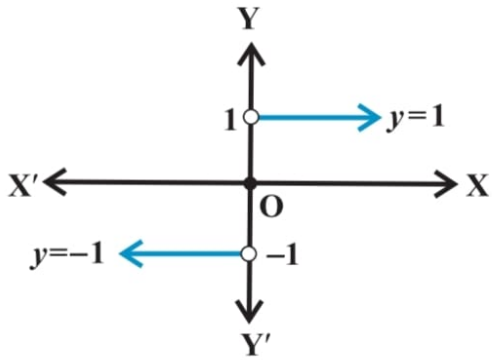

$f(x) = |x|$

**Fig 2.13**

**(vi) Signum function** The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x > 0, \\ 0, & x = 0, \\ -1, & x < 0 \end{cases}$$

is called the *signum function*. The domain of the signum function is $\mathbb{R}$ and the range is the set $\{-1, 0, 1\}$. The graph of the signum function is given by Fig. 2.14.




$f(x) = \frac{|x|}{x}, x$ ‘ 0 and 0 for $x = 0$

Fig 2.14

**(vii) Greatest integer function**

The function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = [x],\ x \in \mathbb{R}$ assumes the value of the greatest integer, less than or equal to $x$. Such a function is called the *greatest integer function.*

From the definition of $[x]$, we can see that

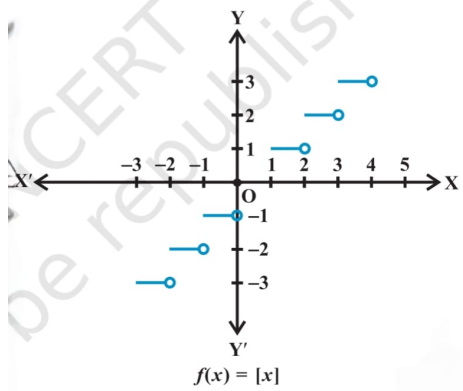
$$[x] = -1 \quad \text{for } -1 \le x < 0$$

$$[x] = \quad 0 \quad \text{for } 0 \le x < 1$$

$$[x] = \quad 1 \quad \text{for } 1 \le x < 2$$

$$[x] = \quad 2 \quad \text{for } 2 \le x < 3 \text{ and so on.}$$

The graph of the function is shown in Fig. 2.15.




$f(x) = [x]$

Fig 2.15

**Fig 2.15**

### 2.4.2 Algebra of real functions

In this Section, we shall learn how to add two real functions, subtract a real function from another, multiply a real function by a scalar (here by a scalar we mean a real number), multiply two real functions and divide one real function by another.

**(i) Addition of two real functions** Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be any two real functions, where $X \subset \mathbb{R}$. Then, we define $(f + g) : X \to \mathbb{R}$ by

$(f + g)(x) = f(x) + g(x), \quad$ for all $x \in X$.

**(ii) Subtraction of a real function from another** Let $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ be any two real functions, where $X \subset \mathbb{R}$. Then, we define $(f - g) : X \to \mathbb{R}$ by

$$(f - g)(x) = f(x) - g(x), \quad \text{for all } x \in X.$$

**(iii) Multiplication by a scalar** Let $f : X \to \mathbb{R}$ be a real valued function and $\alpha$ be a scalar. Here by scalar, we mean a real number. Then the product $\alpha f$ is a function from $X$ to $\mathbb{R}$ defined by

$$(\alpha f)(x) = \alpha f(x), \quad x \in X.$$

**(iv) Multiplication of two real functions** The product (or multiplication) of two real functions $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$ is a function $fg : X \to \mathbb{R}$ defined by

$$(fg)(x) = f(x)g(x), \quad \text{for all } x \in X.$$

This is also called *pointwise multiplication.*

**(v) Quotient of two real functions** Let $f$ and $g$ be two real functions defined from $X \to \mathbb{R}$, where $X \subset \mathbb{R}$. The quotient of $f$ by $g$ denoted by $\dfrac{f}{g}$ is a function defined by

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad \text{provided } g(x) \neq 0,\ x \in X.$$

**Example 16** Let $f(x) = x^2$ and $g(x) = 2x+1$ be two real functions. Find $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$ and $\left(\dfrac{f}{g}\right)(x)$.

**Solution** We have,

$$(f+g)(x) = x^2 + 2x + 1, \quad (f-g)(x) = x^2 - 2x - 1,$$

$$(fg)(x) = x^2(2x+1) = 2x^3 + x^2, \quad \left(\frac{f}{g}\right)(x) = \frac{x^2}{2x+1}, \quad x \neq -\frac{1}{2}.$$

**Example 17** Let $f(x) = \sqrt{x}$ and $g(x) = x$ be two functions defined over the set of non-negative real numbers. Find $(f+g)(x)$, $(f-g)(x)$, $(fg)(x)$ and $\left(\dfrac{f}{g}\right)(x)$.

**Solution** We have,

$$(f+g)(x) = \sqrt{x} + x, \quad (f-g)(x) = \sqrt{x} - x,$$

$$(fg)(x) = \sqrt{x}\,x = x^{3/2}, \quad \left(\frac{f}{g}\right)(x) = \frac{\sqrt{x}}{x} = x^{-1/2}, \quad x \neq 0.$$

## EXERCISE 2.3

1. Which of the following relations are functions? Give reasons. If it is a function, determine its domain and range.
   (i) $\{(2,1),(5,1),(8,1),(11,1),(14,1),(17,1)\}$
   (ii) $\{(2,1),(4,2),(6,3),(8,4),(10,5),(12,6),(14,7)\}$
   (iii) $\{(1,3),(1,5),(2,5)\}$

2. Find the domain and range of the following real functions:
   (i) $f(x) = -|x|$ (ii) $f(x) = \sqrt{9-x^2}$

3. A function $f$ is defined by $f(x) = 2x - 5$. Write down the values of
   (i) $f(0)$ (ii) $f(7)$ (iii) $f(-3)$

4. The function $t$ which maps temperature in degree Celsius into temperature in degree Fahrenheit is defined by
   $$t(C) = \frac{9C}{5} + 32.$$
   Find (i) $t(0)$ (ii) $t(28)$ (iii) $t(-10)$ (iv) The value of $C$, when $t(C) = 212$

5. Find the range of each of the following functions:
   (i) $f(x) = 2 - 3x,\ x \in \mathbb{R},\ x > 0$
   (ii) $f(x) = x^2 + 2$, $x$ is a real number
   (iii) $f(x) = x$, $x$ is a real number

## Miscellaneous Examples

**Example 18**
Let $\mathbb{R}$ be the set of real numbers. Define the real function
$$f : \mathbb{R} \to \mathbb{R} \quad \text{by} \quad f(x) = x + 10$$
and sketch the graph of this function.

**Solution** Here $f(0) = 10$, $f(1) = 11$, $f(2) = 12$, ..., $f(10) = 20$, etc., and $f(-1) = 9$, $f(-2) = 8$, ..., $f(-10) = 0$ and so on.

Therefore, the shape of the graph of the given function assumes the form as shown in Fig. 2.16.

**Remark** The function $f$ defined by $f(x) = mx + c$, $x \in \mathbb{R}$, is called a *linear function*, where $m$ and $c$ are constants. Above function is an example of a linear function.

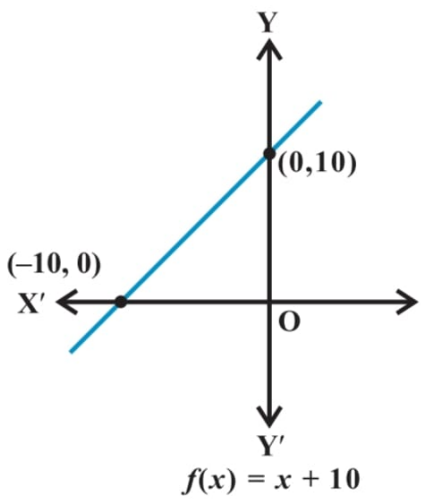


Fig 2.16

**Example 19** Let $R$ be a relation from $\mathbb{Q}$ to $\mathbb{Q}$ defined by
$$R = \{(a,b) : a, b \in \mathbb{Q} \text{ and } a - b \in \mathbb{Z}\}.$$
Show that

1. $(a, a) \in R$ for all $a \in \mathbb{Q}$
2. $(a, b) \in R$ implies that $(b, a) \in R$
3. $(a, b) \in R$ and $(b, c) \in R$ implies that $(a, c) \in R$

**Solution**

1. Since, $a - a = 0 \in \mathbb{Z}$, it follows that $(a, a) \in R$.
2. $(a, b) \in R$ implies that $a - b \in \mathbb{Z}$. So, $b - a \in \mathbb{Z}$. Therefore, $(b, a) \in R$.
3. $(a, b)$ and $(b, c) \in R$ implies that $a - b \in \mathbb{Z}$. $b - c \in \mathbb{Z}$. So,
$$a - c = (a - b) + (b - c) \in \mathbb{Z}.$$
Therefore, $(a, c) \in R$.

**Example 20** Let $f = \{(1, 1), (2, 3), (0, -1), (-1, -3)\}$ be a linear function from $\mathbb{Z}$ into $\mathbb{Z}$. Find $f(x)$.

**Solution** Since $f$ is a linear function, $f(x) = mx + c$. Also, since $(1, 1), (0, -1) \in R$,
$$f(1) = m + c = 1 \quad \text{and} \quad f(0) = c = -1.$$
This gives $m = 2$ and $f(x) = 2x - 1$.

**Example 21** Find the domain of the function
$$f(x) = \frac{x^2 + 3x + 5}{x^2 - 5x + 4}.$$

**Solution** Since $x^2 - 5x + 4 = (x - 4)(x - 1)$, the function $f$ is defined for all real numbers except at $x = 4$ and $x = 1$. Hence the domain of $f$ is
$$\mathbb{R} \setminus \{1, 4\}.$$

**Example 22** The function $f$ is defined by
$$f(x) = \begin{cases} 1 - x, & x < 0, \\ 1, & x = 0, \\ x + 1, & x > 0. \end{cases}$$
Draw the graph of $f(x)$.

**Solution**

Here, $f(x) = 1 - x$, $x < 0$, this gives
$$\begin{aligned} f(-4) &= 1 - (-4) = 5; \\ f(-3) &= 1 - (-3) = 4; \\ f(-2) &= 1 - (-2) = 3; \\ f(-1) &= 1 - (-1) = 2; \text{ etc.,} \end{aligned}$$
and
$$f(1) = 2, \quad f(2) = 3, \quad f(3) = 4,$$
$$f(4) = 5 \text{ and so on for } f(x) = x + 1,\ x > 0.$$
Thus, the graph of $f$ is as shown in Fig. 2.17.

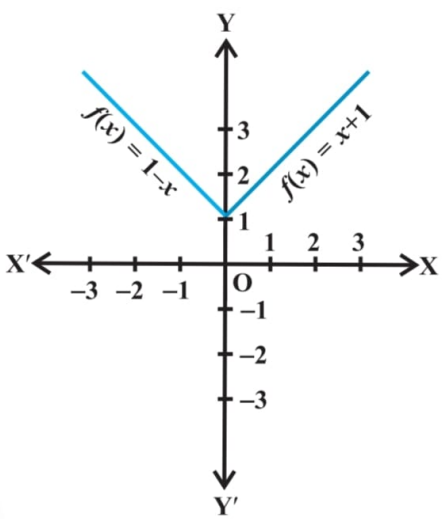


Fig 2.17

## *Miscellaneous Exercise on Chapter 2*

1. The relation $f$ is defined by

$$f(x) = \begin{cases} x^2, & 0 \le x \le 3, \\ 3x, & 3 < x \le 10. \end{cases}$$

   The relation $g$ is defined by

$$g(x) = \begin{cases} x^2, & 0 \le x \le 2, \\ 3x, & 2 \le x \le 10. \end{cases}$$

   Show that $f$ is a function and $g$ is not a function.

2. If $f(x) = x^2$, find

$$\frac{f(1.1) - f(1)}{(1.1 - 1)}.$$

3. Find the domain of the function

$$f(x) = \frac{x^2 + 2x + 1}{x^2 - 8x + 12}.$$

4. Find the domain and the range of the real function $f$ defined by

$$f(x) = \sqrt{x - 1}.$$

5. Find the domain and the range of the real function $f$ defined by

$$f(x) = |x - 1|.$$

6. Let

$$f = \left\{ \left( x, \frac{x^2}{1 + x^2} \right) : x \in \mathbb{R} \right\}$$

   be a function from $\mathbb{R}$ into $\mathbb{R}$. Determine the range of $f$.

7. Let $f, g : \mathbb{R} \to \mathbb{R}$ be defined, respectively, by

$$f(x) = x + 1, \quad g(x) = 2x - 3.$$

   Find $f + g$, $f - g$ and $\dfrac{f}{g}$.

8. Let

$$f = \{(1, 1), (2, 3), (0, -1), (-1, -3)\}$$

   be a function from $\mathbb{Z}$ to $\mathbb{Z}$ defined by

$$f(x) = ax + b,$$

   for some integers $a, b$. Determine $a$ and $b$.

9. Let $R$ be a relation from $\mathbb{N}$ to $\mathbb{N}$ defined by

$$R = \{(a, b) : a, b \in \mathbb{N} \text{ and } a = b^2\}.$$

   Are the following true?

(a) $(a, a) \in R$, for all $a \in \mathbb{N}$
(b) $(a, b) \in R$ implies $(b, a) \in R$
(c) $(a, b) \in R$, $(b, c) \in R$ implies $(a, c) \in R$

Justify your answer in each case.

10. Let $A = \{1, 2, 3, 4\}$, $B = \{1, 5, 9, 11, 15, 16\}$ and

$$f = \{(1, 5), (2, 9), (3, 1), (4, 5), (2, 11)\}.$$

Are the following true?

(a) $f$ is a relation from $A$ to $B$
(b) $f$ is a function from $A$ to $B$

Justify your answer in each case.

11. Let $f$ be the subset of $\mathbb{Z} \times \mathbb{Z}$ defined by

$$f = \{(ab, a + b) : a, b \in \mathbb{Z}\}.$$

Is $f$ a function from $\mathbb{Z}$ to $\mathbb{Z}$? Justify your answer.

12. Let $A = \{9, 10, 11, 12, 13\}$ and let $f : A \to \mathbb{N}$ be defined by

$$f(n) = \text{the highest prime factor of } n.$$

Find the range of $f$.

## Summary

In this Chapter, we studied about relations and functions. The main features of this Chapter are as follows:

- **Ordered pair** A pair of elements grouped together in a particular order.
- **Cartesian product** $A \times B$ of two sets $A$ and $B$ is given by $A \times B = \{(a, b) : a \in A, b \in B\}$.

  In particular $R \times R = \{(x, y) : x, y \in R\}$

  and $R \times R \times R = \{(x, y, z) : x, y, z \in R\}$.
- If $(a, b) = (x, y)$, then $a = x$ and $b = y$.
- If $n(A) = p$ and $n(B) = q$, then $n(A \times B) = pq$.
- $A \times \phi = \phi$.
- In general, $A \times B \neq B \times A$.
- **Relation** A relation $R$ from a set $A$ to a set $B$ is a subset of the cartesian product $A \times B$ obtained by describing a relationship between the first element $x$ and the second element $y$ of the ordered pairs in $A \times B$.
- The *image* of an element $x$ under a relation $R$ is given by $y$, where $(x, y) \in R$.
- The *domain* of $R$ is the set of all first elements of the ordered pairs in a relation $R$.

- The *range* of the relation $R$ is the set of all second elements of the ordered pairs in a relation $R$.
- **Function** A function $f$ from a set $A$ to a set $B$ is a specific type of relation for which every element $x$ of set $A$ has one and only one image $y$ in set $B$. We write $f : A \to B$, where $f(x) = y$.
- $A$ is the domain and $B$ is the codomain of $f$.
- The range of the function is the set of images.
- A real function has the set of real numbers or one of its subsets both as its domain and as its range.
- ***Algebra of functions*** For functions $f : X \to \mathbb{R}$ and $g : X \to \mathbb{R}$, we have

$$\begin{aligned}
(f+g)(x) &= f(x) + g(x), && x \in X,\\
(f-g)(x) &= f(x) - g(x), && x \in X,\\
(f \cdot g)(x) &= f(x) \cdot g(x), && x \in X,\\
(kf)(x) &= k(f(x)), && x \in X, \text{ where } k \text{ is a real number},\\
\left(\frac{f}{g}\right)(x) &= \frac{f(x)}{g(x)}, && x \in X,\ g(x) \neq 0.
\end{aligned}$$

### *Historical Note*

The word **FUNCTION** first appears in a Latin manuscript *"Methodus tangentium inversa, seu de functionibus"* written by Gottfried Wilhelm Leibnitz (1646–1716) in 1673; Leibnitz used the word in the non-analytical sense. He considered a function in terms of "mathematical job" — the "employee" being just a curve.

On July 5, 1698, Johan Bernoulli, in a letter to Leibnitz, for the first time deliberately assigned a specialised use of the term *function* in the analytical sense. At the end of that month, Leibnitz replied showing his approval.

*Function* is found in English in 1779 in Chambers' Cyclopaedia: "The term function is used in algebra, for an analytical expression any way compounded of a variable quantity, and of numbers, or constant quantities".